

# SmartHub Bid API — Reference

The SmartHub serving API exposes the bid-recommendation model over HTTP. It has two functional endpoints plus a health check:

| Method | Path           | Purpose                                                               |
|--------|----------------|-----------------------------------------------------------------------|
| POST   | /recommend_bid | Return the expected-profit-maximizing bid for one lead.               |
| POST   | /explain_bid   | Explain an already-logged prediction (SHAP + optional LLM narrative). |
| GET    | /health        | Liveness + which model is loaded.                                     |

- **Base URL (staging):** `http://<host>:8000` (served via nginx → uvicorn).
- **Content type:** `application/json` for all `POST` requests.
- **Auth:** none at the edge today (network-restricted).
- **Every prediction is logged** server-side to `smarthub_prediction_log` with a unique `prediction_id`, which is returned in the response. SHAP is attached to that row asynchronously (see `/explain_bid`). See `docs/PREDICTION_LOG_SCHEMA.md` for the log schema.

## 1. POST /recommend\_bid

Runs one lead through the bidding policy ( `decide_bid` ) and returns the recommended bid. The response is **lean by default**; pass `verbose: true` to also receive the full decision payload used by the logging schema.

### 1.1 Request body ( `BidRequest` )

**Required parameters** — the request is rejected (HTTP 422) if any is missing. A lead that cannot supply all of these is treated as an upstream data-quality issue for SmartHub to fix, not a case the model should attempt to score.

| Field                         | Type     | Constraints                                | Description                                                                                                                                                                       |
|-------------------------------|----------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>expected_revenue</code> | number   | <code>&gt; 0</code>                        | Reject-discounted expected revenue if the lead is won. Drives the profit calc.                                                                                                    |
| <code>lead_type_id</code>     | integer  | <code>6</code> or <code>1</code>           | Which lead-type model to use ( <code>6</code> =auto, <code>1</code> =home). <b>No default — must be supplied.</b>                                                                 |
| <code>campaign_id</code>      | integer  |                                            | Campaign the lead belongs to.                                                                                                                                                     |
| <code>source_type_id</code>   | integer  |                                            | Lead source / sub-source id (the warehouse source-step / <code>sol_step</code> id).                                                                                               |
| <code>traffic_tier</code>     | string   |                                            | Traffic-quality tier for the lead. Required — different sources have distinct win-rate/winning-bid curves.                                                                        |
| <code>state</code>            | string   |                                            | US state of the lead.                                                                                                                                                             |
| <code>created_at</code>       | datetime | <code>YYYY-MM-DD HH:MM:SS</code> (Pacific) | Date and time the lead was created. The model's timing features ( <code>created_hour</code> , <code>created_day</code> , <code>is_workday</code> ) are <b>derived from this</b> . |

**Optimizer controls** (optional; sensible defaults)

| Field                  | Type   | Default           | Constraints    | Description                                        |
|------------------------|--------|-------------------|----------------|----------------------------------------------------|
| <code>target_cm</code> | number | <code>0.25</code> | $0 \leq x < 1$ | Target contribution margin (sets the bid ceiling). |
| <code>min_bid</code>   | number | <code>0.25</code> | $\geq 0$       | Partner-side minimum / lowest candidate bid.       |
| <code>bid_step</code>  | number | <code>0.25</code> | $> 0$          | Increment between candidate bids in the sweep.     |

**Optional context** (logging / traceability only — never used in the decision)

| Field                     | Type    | Default            | Description                                                                                                                                                     |
|---------------------------|---------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>account_id</code>   | integer | <code>null</code>  | Partner/account the lead belongs to. A grouping parent of <code>campaign_id</code> (which carries the predictive signal), so it adds no model value on its own. |
| <code>lead_ping_id</code> | integer | <code>null</code>  | Correlation key back to <code>public.lead_pings.id</code> , for logging / joining realized outcomes.                                                            |
| <code>verbose</code>      | boolean | <code>false</code> | When <code>true</code> , return the full decision payload (§1.3). Not a feature — never affects the bid.                                                        |

**Optional lead features** (default `null` ; missing values are imputed by the model pipeline)

| Field                             | Type   | Field                                   | Type   |
|-----------------------------------|--------|-----------------------------------------|--------|
| <code>insured</code>              | string | <code>num_vehicles</code>               | number |
| <code>age</code>                  | number | <code>num_drivers</code>                | number |
| <code>home_owner</code>           | string | <code>num_auto_violations</code>        | number |
| <code>dui</code>                  | string | <code>num_auto_accidents</code>         | number |
| <code>sr22_required</code>        | string | <code>num_home_claims</code>            | number |
| <code>gender</code>               | string | <code>continuous_coverage_months</code> | number |
| <code>marital_status</code>       | string | <code>home_property_type</code>         | string |
| <code>military_affiliation</code> | string | <code>current_carrier</code>            | string |

**Required vs optional (updated 2026-08-25, per DS sync).** The required set is exactly the universal request-context fields above — `expected_revenue`, `lead_type_id`, `campaign_id`, `source_type_id`, `traffic_tier`, `state`, and `created_at`. `traffic_tier` is required because different sources have distinct win-rate/winning-bid curves. Every other field is **optional**: - **Logging-only identifiers** (`account_id`, `lead_ping_id`) are never predictive and stay optional so a missing one can't block a bid. - **Lead attributes** (`insured`, `age`, `home_owner`, `num_vehicles`, `num_auto_accidents`, `dui`, `sr22_required`, ...) are imputed by the model pipeline when missing, so the lead is still scored rather than rejected — they inform the model when present but are not required. (`insured` is a model feature but the upstream API already guarantees it, so it need not be mandatory here.)

This matches the implemented validator (registry `mandatory` flags). Any field not listed above is ignored.

**Why request context is passed explicitly rather than looked up.** The decision needs campaign / source / lead-type / timing at request time; resolving them from `lead_ping_id` would require a warehouse lookup that adds ~1 s to the turnaround, which the  $\leq 1$  s SLA can't absorb.

## Example request

```
curl -s -X POST http://localhost:8000/recommend_bid \
-H 'Content-Type: application/json' \
-d '{
  "expected_revenue": 100,
  "campaign_id": 40088,
  "lead_type_id": 6,
  "account_id": 221,
  "source_type_id": 574,
  "lead_ping_id": 998877,
  "created_at": "2026-08-14 14:23:05",
  "home_owner": "true",
  "insured": "true",
  "age": 34,
  "num_vehicles": 2,
  "num_auto_accidents": 0,
  "dui": "false",
  "sr22_required": "false",
  "traffic_tier": "tier_2",
  "state": "CA"
}'
```

## 1.2 Response — default (lean)

Returned when `verbose` is omitted or `false`. Optimized for the hot path.

```
{
  "recommended_bid": 10.25,
  "recommended_bid_predicted_win_rate": 0.9715,
  "recommended_bid_predicted_profit": 87.19,
  "max_bid": 75.0,
  "n_candidate_bids": 300,
  "decision_path": "model",
  "decision_reason": "Standard profit-maximizing bid from the currently-serving model.",
  "model_data_age_days": 0,
  "prediction_id": "d0e044ab-2ba8-4454-88ad-19331f4560a9",
  "lead_ping_id": null
}
```

| Field                                           | Type           | Description                                                                                                                                 |
|-------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <code>recommended_bid</code>                    | number   null  | The chosen bid. <code>null</code> = <b>no viable bid</b> (revenue too low to bid at/above the floor while meeting <code>target_cm</code> ). |
| <code>recommended_bid_predicted_win_rate</code> | number   null  | Model-predicted win probability at the recommended bid.                                                                                     |
| <code>recommended_bid_predicted_profit</code>   | number   null  | Predicted profit at the recommended bid ( $\text{win\_rate} \times (\text{expected\_revenue} - \text{bid})$ ).                              |
| <code>max_bid</code>                            | number         | CM-respecting bid ceiling used for the sweep.                                                                                               |
| <code>n_candidate_bids</code>                   | integer        | Number of candidate bids evaluated by the optimizer.                                                                                        |
| <code>decision_path</code>                      | string         | <code>model</code>   <code>cold_start_fallback</code>   <code>exploration</code> (see §1.4).                                                |
| <code>decision_reason</code>                    | string         | Plain-English explanation of the path taken.                                                                                                |
| <code>model_data_age_days</code>                | integer   null | Age (days) of the serving model's training data; flags staleness (informational).                                                           |
| <code>prediction_id</code>                      | string (uuid4) | Receipt for this prediction's log row; use it with <code>/explain_bid</code> .                                                              |
| <code>lead_ping_id</code>                       | integer   null | Echoed back exactly as sent.                                                                                                                |

### 1.3 Response — verbose ( `verbose: true` )

Includes **all lean fields above**, plus the full logging-schema payload below. The data is packaged from values already computed during the prediction (no extra model inference), so it doesn't change the decision — only the response size. The candidate sweep is **capped to 20** on the wire (the selected bid + the first 19 by bid); the full sweep is still persisted to the log. **SHAP is not included** — retrieve it later via `/explain_bid` using `prediction_id`.

Additional fields:

| Field                                                                                                    | Type                 | Description                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>served_at</code>                                                                                   | string<br>(ISO-8601) | When the prediction was served (UTC).                                                                                                                               |
| <code>tat_seconds</code>                                                                                 | number               | Server turnaround time in seconds.                                                                                                                                  |
| <code>package_version</code>                                                                             | string               | SmartHub package version (semver) that served the row.                                                                                                              |
| <code>lead_type_id</code> / <code>lead_type_name</code>                                                  | integer /<br>string  | Which lead-type model answered (e.g. <code>6</code> / <code>"auto"</code> ).                                                                                        |
| <code>campaign_id</code> / <code>account_id</code> /<br><code>source_type_id</code>                      | integer   null       | Business context echoed back.                                                                                                                                       |
| <code>input_features</code>                                                                              | object               | Raw request snapshot (the lead features as received).                                                                                                               |
| <code>model_input_features</code>                                                                        | object               | The exact row the model scored (post feature-engineering, imputed).                                                                                                 |
| <code>feature_cols</code>                                                                                | array[string]        | Ordered feature columns the serving model used.                                                                                                                     |
| <code>expected_revenue</code> / <code>target_cm</code> /<br><code>min_bid</code> / <code>bid_step</code> | number               | Optimizer inputs echoed back.                                                                                                                                       |
| <code>candidate_bid_generation</code>                                                                    | object               | How the candidate set was built: <code>{method, min_bid, max_bid, bid_step, n_candidates}</code> .                                                                  |
| <code>model_name</code>                                                                                  | string               | Human-friendly production slug (e.g. <code>auto_v8</code> ).                                                                                                        |
| <code>model_version</code>                                                                               | string               | Exact training-run id (e.g. <code>run_20260817T105906Z_2b0f9075</code> ).                                                                                           |
| <code>model_uri</code>                                                                                   | string               | Path/URI of the loaded artifact.                                                                                                                                    |
| <code>model_type</code>                                                                                  | string               | e.g. <code>lightgbm</code> .                                                                                                                                        |
| <code>model_calibrated</code>                                                                            | boolean              | Whether probability calibration is applied.                                                                                                                         |
| <code>training_table_version</code>                                                                      | string               | Training-table snapshot the model came from.                                                                                                                        |
| <code>model_data_min_created_at</code> /<br><code>model_data_max_created_at</code>                       | string               | Date range of the training data.                                                                                                                                    |
| <code>recommended_bid_predicted_cm</code>                                                                | number   null        | Predicted contribution margin at the recommended bid ( <code>profit</code> / <code>expected_revenue</code> ).                                                       |
| <code>serving_config</code>                                                                              | object               | Serving-policy snapshot: <code>{exploration_variance_pct, recency_window_days, cold_start_fallback_bid_pct}</code> .                                                |
| <code>candidate_evaluations</code>                                                                       | array[object]        | <b>Capped to ≤20.</b> Each entry: <code>{bid, predicted_win_rate, expected_profit, selected}</code> ; exactly one has <code>selected: true</code> (the chosen bid). |

### Example (verbose, abridged)

```
{
  "recommended_bid": 10.25,
  "recommended_bid_predicted_win_rate": 0.9715,
  "recommended_bid_predicted_profit": 87.19,
  "recommended_bid_predicted_cm": 0.8719,
  "decision_path": "model",
  "prediction_id": "7d12c4aa-9ede-4c29-8fef-cde14190c614",
  "served_at": "2026-08-17T11:13:03.601597+00:00",
  "package_version": "0.1.0",
  "model_name": "auto_v8",
  "model_version": "run_20260817T105906Z_2b0f9075",
  "model_type": "lightgbm",
  "model_calibrated": true,
  "training_table_version": "2026-08-17T083011Z",
  "feature_cols": ["bid", "age", "...", "current_carrier"],
  "input_features": { "campaign_id": 13, "state": "CA", "age": 34 },
  "model_input_features": { "bid": 0.25, "age": 34.0, "state": "CA" },
  "candidate_bid_generation": {
    "method": "equally_spaced", "min_bid": 0.25, "max_bid": 75.0,
    "bid_step": 0.25, "n_candidates": 300
  },
  "serving_config": {
    "exploration_variance_pct": 0.1, "recency_window_days": 30,
    "cold_start_fallback_bid_pct": 0.5
  },
  "candidate_evaluations": [
    {"bid": 0.25, "predicted_win_rate": 0.0876, "expected_profit": 8.7393, "selected": false},
    "... first 19 by bid ...",
    {"bid": 10.25, "predicted_win_rate": 0.9715, "expected_profit": 87.1932, "selected": true}
  ]
}
```

## 1.4 Decision paths

| decision_path       | Meaning                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model               | Normal case — profit-maximizing bid from the serving model.                                                                                                                                |
| cold_start_fallback | No model promoted yet for this lead type; bids a configured fraction of the way from <code>min_bid</code> to the ceiling. <code>recommended_bid_predicted_*</code> are <code>null</code> . |
| exploration         | Scheduled probe — the model's optimal bid is perturbed $\pm$ <code>exploration_variance_pct</code> on a reproducible hour-of-week schedule.                                                |

## 2. POST /explain\_bid

Explains a **previously logged** prediction by id. It loads the exact model version that served that prediction and computes SHAP on the exact scored row — it **never re-runs the bid decision**, so the explanation always matches the logged bid. The explanation is also persisted back onto the same log row.

## 2.1 Request body ( `ExplainRequest` )

| Field                       | Type    | Required                 | Default           | Description                                                              |
|-----------------------------|---------|--------------------------|-------------------|--------------------------------------------------------------------------|
| <code>prediction_id</code>  | string  | <input type="checkbox"/> | —                 | The <code>prediction_id</code> returned by <code>/recommend_bid</code> . |
| <code>with_llm</code>       | boolean |                          | <code>true</code> | Include a natural-language <code>explanation</code> (LLM narrative).     |
| <code>with_bid_curve</code> | boolean |                          | <code>true</code> | Include the predicted win-rate/profit curve around the bid.              |

### Example request

```
curl -s -X POST http://localhost:8000/explain_bid \
-H 'Content-Type: application/json' \
-d '{"prediction_id": "7d12c4aa-9ede-4c29-8fef-cde14190c614", "with_llm": false}'
```

## 2.2 Response

```
{
  "base_prediction": 0.114,
  "prediction": 0.9715,
  "feature_contributions": [
    {"feature": "num_auto_accidents", "value": 1, "contribution": 0.84},
    {"feature": "age", "value": 34.0, "contribution": -0.31}
  ],
  "top_factors": [
    {"feature": "num_auto_accidents", "value": 1, "shap": 0.84, "direction": "increased"},
    {"feature": "age", "value": 34.0, "shap": -0.31, "direction": "decreased"}
  ],
  "base_win_rate": 0.114,
  "bid_curve": [
    {"bid": 8.0, "predicted_win_rate": 0.93, "expected_profit": 85.5}
  ],
  "explanation": "The recommended bid is driven up mainly by ...",
  "prediction_id": "7d12c4aa-9ede-4c29-8fef-cde14190c614",
  "lead_ping_id": null
}
```

| Field                              | Type           | Description                                                                                                                                                              |
|------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>base_prediction</code>       | number   null  | Model's average win rate before this lead's factors (log-odds base, expressed as a probability).                                                                         |
| <code>prediction</code>            | number   null  | The served (calibrated) win rate for this lead — equals the logged <code>recommended_bid_predicted_win_rate</code> .                                                     |
| <code>feature_contributions</code> | array[object]  | <b>Every</b> feature's SHAP contribution: <code>{feature, value, contribution}</code> , sorted by <code> contribution </code> desc. Contributions are in log-odds units. |
| <code>top_factors</code>           | array[object]  | Top-N view: <code>{feature, value, shap, direction}</code> ( <code>direction</code> = <code>increased</code> / <code>decreased</code> ).                                 |
| <code>base_win_rate</code>         | number   null  | Same as <code>base_prediction</code> ; kept for existing consumers.                                                                                                      |
| <code>bid_curve</code>             | array[object]  | Only if <code>with_bid_curve=true</code> — predicted metrics across nearby bids.                                                                                         |
| <code>explanation</code>           | string         | Only if <code>with_llm=true</code> — natural-language narrative generated from the SHAP payload.                                                                         |
| <code>prediction_id</code>         | string         | Echoed back.                                                                                                                                                             |
| <code>lead_ping_id</code>          | integer   null | From the logged row.                                                                                                                                                     |

**Note on numbers:** `feature_contributions` + `base_prediction` sum to the model's *uncalibrated* output, so they won't exactly reconstruct the *calibrated* `prediction`. Use contributions for *why* (direction/magnitude) and `prediction` for the served win probability.

**Robustness:** the explanation never 500s — if SHAP can't run (e.g. an unsupported model type or an unloadable artifact), it returns empty `feature_contributions` / `top_factors` with an `explanation` describing why.

### 3. GET /health

Liveness probe and current model status.

```
curl -s http://localhost:8000/health
```

```
{
  "status": "ok",
  "lead_type_id": 6,
  "model_uri": "/app/data/models/auto/run_20260817T105906Z_2b0f9075.pkl",
  "model_loaded": true
}
```

| Field                     | Type          | Description                                                                                                                     |
|---------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>status</code>       | string        | <code>"ok"</code> when the app is up.                                                                                           |
| <code>lead_type_id</code> | integer       | Default lead type checked.                                                                                                      |
| <code>model_uri</code>    | string   null | Resolved artifact path for the default model.                                                                                   |
| <code>model_loaded</code> | boolean       | Whether that model is currently cached in memory. <code>false</code> right after startup until the first request lazy-loads it. |

## 4. Errors

| Status                    | When                                                                                                                                          | Body                                                          |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 422 Unprocessable Entity  | Request fails validation (missing required field, malformed <code>created_at</code> , <code>expected_revenue ≤ 0</code> , wrong type).        | FastAPI validation detail listing the offending fields.       |
| 404 Not Found             | <code>/explain_bid</code> with an unknown <code>prediction_id</code> .                                                                        | <code>{"detail": "No prediction logged for id '...'."}</code> |
| 500 Internal Server Error | Unexpected serving failure (e.g. model artifact fails to load). The failure is logged with a <code>status: "error"</code> prediction-log row. | <code>Internal Server Error</code> .                          |

## 5. Notes & conventions

- **`prediction_id` is a receipt.** `/recommend_bid` generates and returns it before the log row is written (the write is decoupled off the request path), so a lookup by it can rarely be empty if the async insert failed — treat it as "the id we intend to log under."
- **Logging never affects serving.** Prediction-log writes and SHAP enrichment happen off the response path; a logging outage never delays or fails a bid.
- **`verbose` adds no inference** and doesn't change latency on the default (lean) path — only opt-in callers pay the (small) serialization cost.
- **SHAP is asynchronous.** For `/recommend_bid` , SHAP is backfilled onto the log row by a separate worker; to read it on demand, call `/explain_bid` with the `prediction_id` .